

$$\bullet \left\{ \begin{array}{l} \mathrm{d}X_t^i = \mathrm{Pev}(\theta_t^i) \mathrm{d}t + \sqrt{2\sigma_x} \mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i = \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} \mathrm{d}B_t^i \end{array} \right.$$

Well-posedness

$$\circ \text{ Singular (Newtonian) } \nabla K \text{ and } \sigma_x > 0: |\nabla K(x)| \sim |x|^{-1}$$

■ If τ small enough, solutions exist for all N and any time $t \geq 0$

■ And: convergence in law to mean-field limit as $N \rightarrow \infty$ ($L^1 \cap L^\infty$ chaotic initial law)

◦ Via recent compactness method for Keller-Segel (Jourdain & Fournier, Ann. Appl. Prob., 2017) (see dW, thesis, 2025)

○ Compare with moderate, cut-off interactions and entropy solutions (for review see Chainton & Diez, Kinet. Rel. Models, 2022)

Ant stochastic analysis

Ant stochastic analysis

- $$\begin{cases} dX_t^i &= P v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$
- Well-posedness
 - Singular (Newtonian) ∇K and $\sigma_x > 0$: $|\nabla K(x)| \sim |x|^{-1}$
 - **If τ small enough**, solutions exist for all N and any time $t \geq 0$
 - And: convergence in law to mean-field limit as $N \rightarrow \infty$ ($L^1 \cap L^\infty$ chaotic initial law)
 - Via recent compactness method for Keller-Segel (Jourdain & Fournier, Ann. Appl. Prob., 2017) (see dW, thesis, 2025)
 - Compare with moderate, cut-off interactions and entropy solutions (for review see Chaintron & Diez, Kinet. Rel. Models, 2022)

Ant analysis

$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$

- Mean-field limit PDE

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$